10- FRANCK-HERTZ EXPERIMENT

PURPOSE of the EXPERIMENT

The observation of the energy levels of atoms has discrete values. Recording the inverted current (inrush current) intensity I in a Franck-Hertz tube as a function of the anode voltage U. Calculation of the excitation energy E from the positions of the minima or maxima of the current intensity by difference formalization.

GENERAL INFORMATION

The explanation of critical physical phenomena of small particle systems was not possible with the classical mechanical view and this was the main reason for the emergence and establishment of the quantum mechanical view. Unfortunately, many attempts to determine the energy states of the atom using the classical view were unsuccessful, the common feature in all these failed attempts was that the classical view assumed the continuity of energy. In the early 1900s, Max Planck put forward a thesis that was not easy to accept, and which was met with a reaction from many scientific circles. Max Planck argued for the discontinuity of energy. I wonder if this idea would find the value it deserved at the time if it would break the stereotypes of a long period. It was indeed a way of thinking that was not easy to accept. However, it was not very difficult for this idea to be accepted, because many important problems in physics, which had not been solved until then, were perfectly solved within the framework of this view. Among these problems, the black body radiation solved by Planck has a very important place. In addition, Einstein, taking Planck's thesis into account, solved the photoelectric effect in 1905 and made a major contribution to the further consolidation of quantum theory.

J. Frank and G. Hertz were the first researchers to experimentally demonstrate the validity of quantum mechanical thinking and that atoms have discrete energy values. With their experiment on the mercury atom in 1914, these two scientists accomplished a very important feat that proved the validity of quantum theory.

Atoms can only have certain energy values, and these energy values are quantized (discrete) depending on the structure of the atom. These energy values, or in other words energy levels, are a distinctive physical characteristic of the atom. The energy levels above the fundamental energy level of the atom are called excited energy levels or excited energy levels. The energies of the energy levels of each atom are different from those of other atoms.

In our experiment, we aim to verify the above judgment by using the mercury atom. In this framework, our experiment aims to show that the atom will have discrete energy levels and to experimentally find only the first excitation energy of the mercury atom.

Atoms, the building blocks of matter, always prefer to be in the most stable state they can be in. The most stable state of the atom is the state in which the atom is at the fundamental energy level. In various ways, the atom passes from the stable state to a new state with external excitation, this state corresponds to one of the excited states of the atom. The atom is unstable in this excited state and wants to go back to the fundamental level, that is, the stable level. The time the atom can be in the excited state is a very small period, around 10-8 s, which is defined as the lifetime or lifetime of the excited level, after which the atom returns to the ground state. During this return, due to the conservation of energy, the atom emits a photon with an energy equal to the difference between the excited and ground state energies. This emitted photon can propagate at the speed of light in any direction of space.

Ei: Excited level energy and Es: ground-level energy

The energy of the emitted photon: $E_f = E_s - E_i = h\nu_f$

Here ν_f: frequency of the emitted photon, h: Planck's constant.

There are two different methods to excite an atom at the fundamental electronic energy level. Either accelerated electrons or ions are used to excite the atom or atoms are bombarded with photons. Both methods can excite the atom to the desired energy value, but there are significant differences between these two methods.

In this experiment, the mercury atom is excited with accelerated free electrons. When charged particles enter a uniform electric field, they gain potential energy from the electric field. The charged particles will convert this potential energy into kinetic energy during motion. If the charged particles manage to transfer their kinetic energy to the atoms through inelastic collisions with the atoms, they will perform excitation processes.

In the experiment, free electrons are obtained and accelerated in a uniform electric field to collide with mercury atoms in the gas phase. The conditions of the collisions are very important; the number of mercury atoms in the gas phase in the furnace where the collisions take place is an important factor for the probability of collisions and the mean free path, i.e. the distance between two consecutive collision points. In the experimental set, the conditions of the gas

environment were such that mercury atoms were excited only to the first excited energy level throughout the experiment.

In order to obtain a uniform electric field, it is sufficient to apply a direct voltage (D.C. potential difference) between the cathode and the anode. This voltage will be varied throughout the experiment with a potentiometer.

The cathode is heated with the help of a filament behind it to release electrons. These electrons are accelerated under the potential difference between the cathode and the anode. The potential energy of an electron under this accelerating voltage (V) is equal to the product of the electron's charge and the magnitude of the applied voltage.

Potential energy gained by free electrons: $E_p = eV$

During the movement starting from the cathode and continuing towards the anode, the electrons convert their potential energy into kinetic energy, as long as there is no energy loss, the kinetic energy they have when they reach the anode $E_k = eV = ½\ mv^2$

Interactions between free electrons and mercury atoms can take two forms; if the interactions take the form of collisions of the flexible type, the electrons collide with the mercury atoms like two billiard balls colliding with each other and continue their motion without any loss of kinetic energy. In this case, the atoms they collide with also remain at the fundamental electronic level without excitation. We call this type of interaction, which does not change the kinetic energy of the free electrons, elastic interactions. In this case, the electrons reach the anode with a kinetic energy of $E_k = eV$, depending on the voltage applied between the cathode and the anode.

The second possibility is in the form of inelastic collisions. In such a case, some of the free electrons may transfer their kinetic energy to the mercury atoms with which they collide, in which case the free electron loses all or most of its kinetic energy, while the mercury atom with which it collides rises to the first excitation energy level and emits a photon to return to the ground state. The basic condition for such an interaction to occur is that the kinetic energy of the free electrons that will collide with mercury atoms must have a kinetic energy equal to or greater than the first excited energy level of the mercury atom. The free electrons that make such an interaction with mercury atoms either lose all or part of their energy. Therefore, their kinetic energy is either zero or very small compared to other non-colliding electrons. The most important part of the experiment is to distinguish these small kinetic energy electrons from the others and to detect the excitation event. A good solution is to apply a stopping voltage in the opposite direction to the accelerating voltage in order to remove these electrons which have lost

most of their kinetic energy. With the help of this stopping voltage, the free electrons that have lost their kinetic energy, in other words, the free electrons that have succeeded in exciting mercury atoms, can be separated and, accordingly, the excitation regions can be detected.

Free electrons, which have lost most of their kinetic energy, cannot overcome the potential wall created by the applied restraining voltage, but if the accelerating voltage continues to be increased, these electrons stopped at the potential wall may again have sufficient kinetic energy and can overcome this barrier wall.

The most important element of our experimental environment is our furnace with an evacuated discharge tube as shown in Figure 1. The evacuated tube contains a cathode, a perforated anode, a collector electrode, a filament, and an appropriate amount of liquid mercury element at room temperature (25oC). The furnace is heated to a temperature slightly above the vaporization temperature of mercury to provide suitable experimental conditions. The mercury element starts to evaporate at 185°C and the operating temperature is fixed at around 190°C, thus maintaining a constant temperature and keeping the probability of collisions between mercury atoms in the gas phase environment at a constant value, as well as the average free path, which depends on the collisions between free electrons and mercury atoms sent from the cathode into the gas environment.

Another important experimental element in the experimental setup is the rectifier. The uniform electric field, which is required to provide kinetic energy to the free electrons to be used in the process of excitation of mercury atoms and to stop the electrons that can succeed in the excitation process, can only be obtained by applying a uniform voltage. For this reason, this experimental element called a rectifier is used to convert the alternating city voltage into direct voltage. The free electrons that will be used to excite mercury atoms in the experiment are provided by an experimental element called a filament, which is made of a material whose electrons in the outermost orbitals can be easily detached when heated.

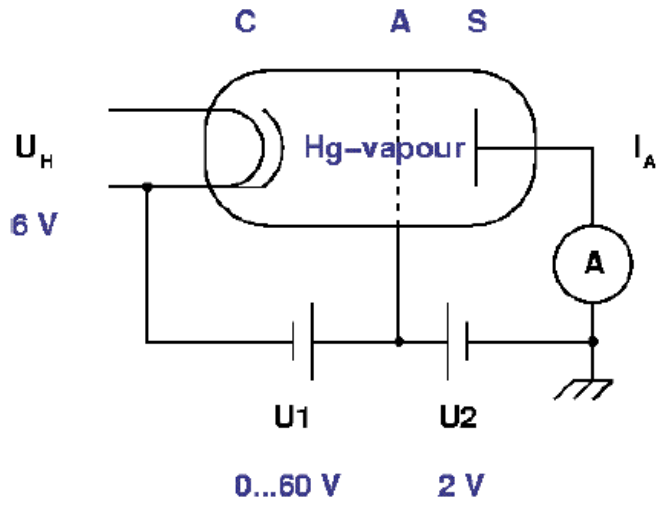


Figure 1: Franck Hertz Tube

With an anode voltage of U1 (4.9 V), the kinetic energy of the electrons is sufficient to bring the valence electron of mercury to the first excitation level 6 $^3P_1$ by inelastic collision. Due to the energy loss, the electron can now no longer travel further in the inverse field between the anode A and the counter electrode S: I current is at a minimum. If the anode voltage is now increased further, the electron kinetic energy is again sufficient to overcome the inverse field: I current intensity increases. When U1 = 4.9 V, two atoms are excited with the same electron because the kinetic energy is high enough: we get a second minimum (Figure 2). This is why the I - U1 graph shows an equal distance between maxima and minima.

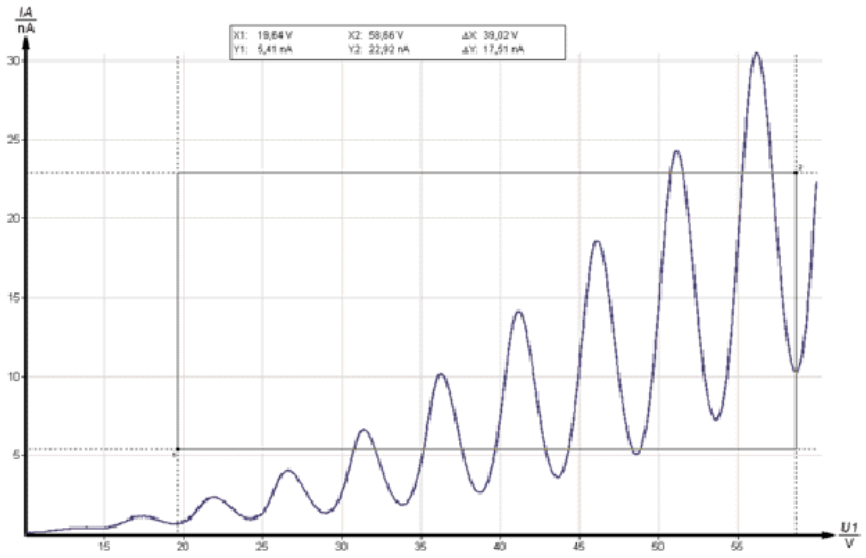


Figure 2: I - U1 graph of the mercury atom.

However, these minima are not well defined because of the initial thermal distribution of electron velocities.

Voltage U1 between anode and cathode $U_1 = U + (\phi_A - \phi_C)$

Here U is the applied voltage, ϕ_A and ϕ_C are the work function voltages of the anode and cathode respectively. Since the excitation energy E is calculated from the voltage differences at minima, the work function voltages are not important here.

According to classical theory, the energy levels that mercury atoms excite can be random. According to quantum theory, however, a defined energy level of the atom appears suddenly.

The trajectory of the I - UA curve was first explained by this and thus a proof of quantum theory was demonstrated.

The excited mercury atom releases the absorbed energy again with the emission of a photon. When the excitation energy E is 4.9 eV, the wavelength of this photon,

$$\lambda = \frac{c\,h}{E} = 253\,nm$$

$$c = 2.9979 \cdot 10^{8}\,\frac{m}{s}$$

$$h = 4.136 \cdot 10^{-15}\,eV$$

This transition is in the UV (ultraviolet) region.

For our calculations, the voltage values of the minima are calculated. By averaging the differences between these values, we find the excitation energy E of the mercury atom.

From the measurements in Figure 2, $E = (4.86 \pm 0.009)eV$

HOW TO CONDUCT THE EXPERIMENT

In this experiment, electrons are accelerated in a tube filled with mercury atoms. The excitation energy of mercury is calculated from the distance between equidistant minima of the electron current in a reversible variable electric field.

A schematic representation of the setup used in the experiment is shown below in Figure 3. The experimental setup consists of a control unit, a Franck-Hertz furnace with an Hg tube, and a computer as shown in Figure 3.



Figure 3: Franck-Hertz experimental setup.

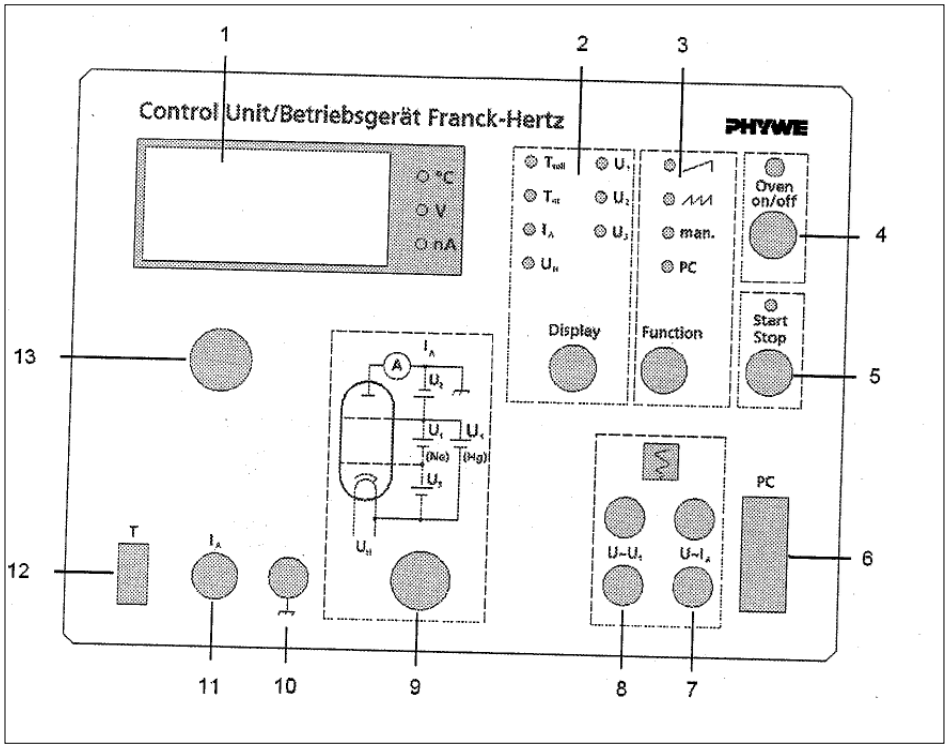


Figure 4: Control unit

Elements of the Control Unit:

1. Three-digit digital display with temperature T, anode current IA, and voltage UH, U1, U2, or U3.

2. "Display" button which allows us to select the attribute to be displayed.

3. "Function" button allowing the selection of "sawtooth", "ramp", "manual control" or "PC control".

4. “Oven on /off” button to activate the heating of the Hg oven.

5. “Start / Stop” button to start or stop the measurement.

6. The socket that serves as the interface for connecting the control unit to the computer.

7. One pair of 4 mm U~IA sockets, Analog output (Y): Voltage proportional to the anode current

8. One pair of 4mm U~IA sockets, Analog output (X): Voltage proportional to the acceleration voltage U1.

9. DIN socket connecting the voltages (UH, U1, U2 and U3) to the tube.

10. GND connector.

11. BNC socket “IA” with input for anode current measurement.

12. Temperature input T. Thermal dissociation socket for DIN plug connection with NiCr-Ni thermal dissociation.

13. Rotary knob for setting the temperature and voltage values.

14. At the rear of the instrument, it provides the voltage for the temperature required for the mercury tube in the Franck-Hertz furnace. Socket with earth connection.

Franck-Hertz Oven:

There is a discharge tube with evacuated air inside. Inside the tube, there is a cathode, a perforated anode, a collector electrode, a filament, and a mercury element at room temperature.

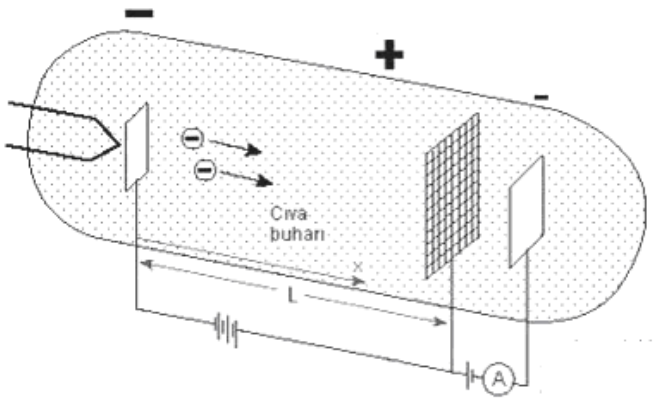


Figure 5: Evacuated Discharge Tube.

Cathode: Produces electrons when heated.

Perforated Anode: Also called grid. It is located opposite the cathode. It is + charged and accelerates the electrons produced.

Collector Electrode: Located behind the Perforated Anode. It collects electrons reaching its surface and creates current. It is slightly negative compared to the perforated anode to ensure that only electrons that can exceed a certain energy level can reach it.

Flame: Made of a material whose electrons in the outermost orbitals can be easily detached when heated, it is the experimental element that generates the free electrons used to excite the mercury atom.

At a certain voltage (U1 = U2), which depends on temperature, a reddish charge-discharge occurs between the anode and cathode by ionization. Therefore, meaningful measurements can only be taken at voltages U1 < U2.

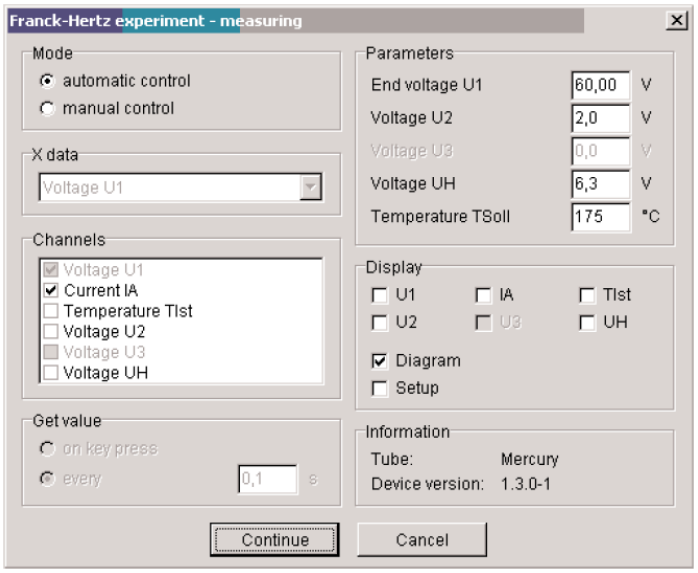


Figure 6: Measurement parameters.

1. Set up the experimental set as shown in Figure 3. Connect the operating unit to the computer.

2. Turn on the control unit and set it to Cobra PC.

3. Start the measurement program and select the Cobra3 Franck-Hertz tester. Enter the parameters given in Figure 6 into the computer and press continue. Now the oven of the Franck-Hertz tube will be heated to 175 ^{0}C and the graph between U1 and IA will be observed as shown in Figure 2.

4. Then enter the U1 values into the computer. Read the IA values and record them in Table 1, draw the graph between U1 and IA by hand, and find the 1st excitation energy of the mercury atom.

5. Find the wavelength of the emitted photon using the excitation energy and determine in which region of the electromagnetic spectrum it lies.

Figure 2: Franck-Hertz curve recorded at T=175 ^{0}C and U2 = 2V.

Table 1: Results

MAX		**MIN**	
U_1 (v)	I_A(nA)	U_1 (v)	I_A(nA)

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